

Research & Development Document

AI-Powered SOC Assistant Integration with Wazuh SIEM

1. Introduction

Security Information and Event Management (SIEM) platforms play a vital role in modern cybersecurity by collecting, correlating, and analyzing security events from endpoints, servers, applications, and network devices. Wazuh is an open-source SIEM solution that provides log analysis, intrusion detection, file integrity monitoring, vulnerability detection, and endpoint security.

Although Wazuh effectively detects security events, security analysts still spend considerable time understanding alerts, determining their severity, and deciding the appropriate response. To reduce manual effort and improve response time, this project focuses on integrating an AI-powered Security Operations Center (SOC) Assistant with Wazuh.

The AI module automatically analyzes security alerts generated by Wazuh, categorizes threats, maps them to the MITRE ATT&CK framework, assigns risk levels and priorities, and recommends remediation actions.

2. Objective

The primary objectives of this research were:

- Study Wazuh architecture and deployment.
- Deploy Wazuh Manager using Docker.
- Configure a Windows endpoint with the Wazuh Agent.
- Verify communication between the agent and manager.
- Develop a Python-based AI monitoring application.
- Perform automated security alert analysis.
- Generate intelligent security recommendations.
- Reduce manual investigation time for SOC analysts.

3. Research on Wazuh

The research phase involved understanding the following Wazuh components:

Wazuh Manager

Processes security logs received from agents, applies detection rules, generates alerts, and coordinates responses.

Wazuh Agent

Installed on Windows endpoints to monitor:

- Windows Event Logs
- System activities
- File integrity
- Running processes
- Security events

Wazuh Indexer

Stores processed alerts and makes them searchable.

Wazuh Dashboard

Provides a web interface for monitoring alerts, agents, and security events.

4. Research on AI Integration

The research focused on identifying how AI could enhance Wazuh by:

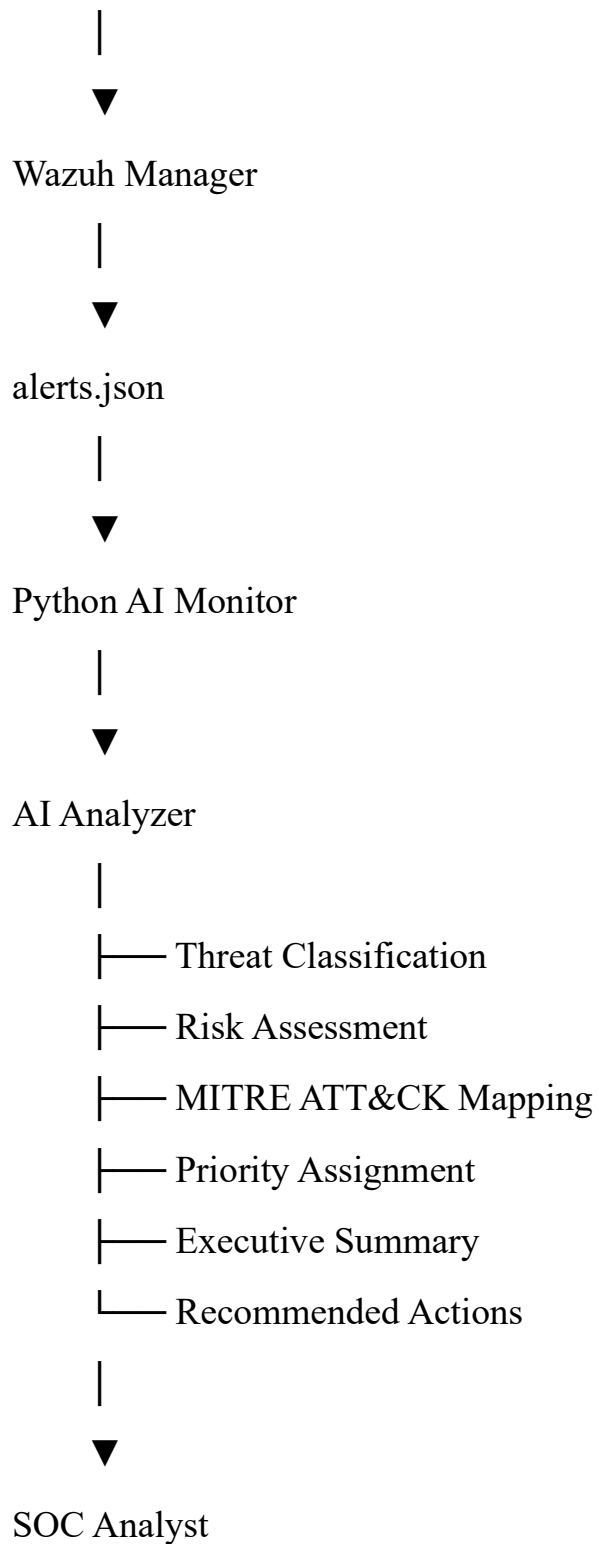
- Explaining alerts in human-readable language.
- Categorizing security events.
- Assigning priorities.
- Mapping alerts to MITRE ATT&CK techniques.
- Providing remediation recommendations.
- Assisting SOC analysts in incident triage.

5. System Architecture

Windows Endpoint



Wazuh Agent



6. Implementation

The implementation consisted of two Python modules.

ai_monitor.py

Responsibilities:

- Continuously monitor Wazuh alerts.
- Read new alerts from alerts.json.
- Parse JSON data.
- Forward alerts to the AI analyzer.
- Display results in real time.

ai_analyzer.py

Responsibilities:

- Determine threat category.
- Assign risk level.
- Generate incident priority.
- Map alerts to MITRE ATT&CK.
- Produce an executive summary.
- Recommend remediation actions.
- Display AI confidence score.

7. Features Implemented

The AI assistant currently supports:

- Real-time alert monitoring
- Threat categorization
- Risk assessment
- Priority assignment (P1–P4)
- MITRE ATT&CK mapping
- Executive summaries
- Security recommendations
- AI confidence score generation
- Continuous monitoring of new alerts

8. Testing Performed

The implementation was validated using multiple scenarios:

Test 1 – Windows Successful Logon

- Alert received successfully.
- AI categorized the event.
- Risk level assigned.
- Recommendations generated.

Test 2 – Failed Login Attempt

- Authentication alert detected.
- AI processed the event.
- Threat classification verified.

Test 3 – Continuous Monitoring

- New alerts detected automatically.
- AI generated analysis without manual execution.

Test 4 – Windows Security Events

Validated processing of multiple Windows Event IDs to ensure reliable monitoring and analysis.

9. Results

The developed solution successfully:

- Established communication between the Windows endpoint and Wazuh Manager.
- Monitored Windows security events in real time.
- Parsed Wazuh alerts.
- Generated AI-assisted threat analysis.
- Produced actionable security recommendations.
- Demonstrated automated SOC alert enrichment.

10. Challenges Faced

Docker Deployment

Initially deployed the latest Wazuh Docker repository, which used an unsupported version for the project. The deployment was recreated using Wazuh 4.12.

Certificate Generation

Docker deployment required SSL certificates before services could start. Certificates were generated using the official Wazuh certificate generation process.

Windows Agent Installation

The Wazuh service was not created correctly during the first installation. The issue was resolved by removing the incomplete installation, reinstalling the agent, authenticating it with the manager, and restarting the service.

Agent Registration

The Windows endpoint did not initially appear in the Wazuh dashboard. After verifying the manager IP and authenticating the agent using agent-auth, the connection was established successfully.

Accessing Wazuh Alerts

Since the Wazuh Manager was deployed in Docker, the alert file was not directly accessible from the host. Docker commands were used to retrieve alerts from inside the container.

Python Development Issues

Several syntax errors, indentation errors, and JSON parsing issues occurred during development. Exception handling and improved monitoring logic were implemented to ensure stable execution.

OpenAI API Integration

During AI integration, authentication (401) and quota (429) errors were encountered due to API configuration and usage limits. API connectivity was successfully verified. To ensure uninterrupted project completion, a local rule-based AI analyzer was implemented while keeping the architecture ready for future LLM integration.

Continuous Monitoring

Reading the latest alert occasionally resulted in incomplete JSON objects. The monitoring logic was enhanced to skip partial entries and continue processing without interruption.

11. Advantages

- Faster incident analysis.
- Reduced manual effort.
- Intelligent alert enrichment.
- Modular architecture.
- Easy integration with additional AI models.
- Improved SOC efficiency.
- Scalable monitoring solution.

12. Limitations

- Current AI logic is primarily rule-based.
- Limited threat categories are implemented.
- Duplicate Windows events require filtering.
- External threat intelligence is not yet integrated.

13. Future Enhancements

Future improvements identified during the research include:

- Integration with GPT-5 or local LLMs.
- VirusTotal integration.
- AbuseIPDB integration.
- Automated incident report generation.
- Slack and Microsoft Teams notifications.
- Email alerting.
- Historical trend analysis.
- Threat intelligence enrichment.

- Dashboard-based AI visualization.
- Machine learning-based anomaly detection.

14. Conclusion

The research successfully demonstrated the integration of an AI-powered Security Operations Center (SOC) Assistant with Wazuh SIEM. The developed solution continuously monitors Windows security events, analyzes Wazuh alerts, classifies threats, maps alerts to the MITRE ATT&CK framework, assigns risk levels and priorities, and generates actionable remediation recommendations.

The implementation demonstrates how AI can assist SOC analysts by reducing manual effort, improving alert triage, and accelerating incident response. The modular design also provides a strong foundation for future integration with advanced Large Language Models (LLMs) and external threat intelligence platforms.